



MDPs: Q-learning

→ Opt বেব করার try করি



on each (s, a, r, s')

$$\hat{V}_{opt}(s') = \max_{a' \in \text{Actions}(s')} \hat{Q}_{opt}(s', a')$$



expected utility starting from state s'

(কিন্তু $s \neq s'$ লাগে না ?)

→ over all possible action $\hat{V}_{opt}(s')$ বের করার try করছি

Q-learning

Problem: model-free Monte Carlo and SARSA only estimate Q_π , but want Q_{opt} to act optimally

Output	MDP	reinforcement learning
Q_π	policy evaluation	model-free Monte Carlo, SARSA
Q_{opt}	value iteration	Q-learning

- Recall our goal is to get an optimal policy, which means estimating Q_{opt} .
- The situation is as follows: Our two methods (model-free Monte Carlo and SARSA) are model-free, but only produce estimates Q_{π} . We have one algorithm, model-based value iteration, which can be used to produce estimates of Q_{opt} , but is model-based. Can we get an estimate of Q_{opt} in a model-free manner?
- The answer is yes, and Q-learning is an algorithm that accomplishes this.
- One can draw an analogy between reinforcement learning algorithms and the classic MDP algorithms. MDP algorithms are offline, RL algorithms are online. In both cases, algorithms either output the Q-values for a fixed policy or the optimal Q-values.

Q-learning

Bellman optimality equation:

$$Q_{\text{opt}}(s, a) = \sum_{s'} T(s, a, s') [\text{Reward}(s, a, s') + \gamma V_{\text{opt}}(s')]$$



Algorithm: Q-learning [Watkins/Dayan, 1992]

On each (s, a, r, s') :

$$\hat{Q}_{\text{opt}}(s, a) \leftarrow (1 - \eta) \underbrace{\hat{Q}_{\text{opt}}(s, a)}_{\text{prediction}} + \eta \underbrace{(r + \gamma \hat{V}_{\text{opt}}(s'))}_{\text{target}}$$

Recall: $\hat{V}_{\text{opt}}(s') = \max_{a' \in \text{Actions}(s')} \hat{Q}_{\text{opt}}(s', a')$

- To derive Q-learning, it is instructive to look back at the Bellman optimality equation for Q_{opt} . There are several changes that take us from this recurrence to Q-learning. First, we don't have an expectation over s' , but only have one sample s' .
- Second, because of this, we don't want to just replace $\hat{Q}_{\text{opt}}(s, a)$ with the target value, but want to interpolate between the old value (prediction) and the new value (target).
- Third, we replace the actual reward $\text{Reward}(s, a, s')$ with the observed reward r (when the reward function is deterministic, the two are the same).
- Finally, we replace $V_{\text{opt}}(s')$ with our current estimate $\hat{V}_{\text{opt}}(s')$.
- Importantly, the estimated optimal value $\hat{V}_{\text{opt}}(s')$ involves a maximum over actions rather than taking the action of the policy. This max over a' rather than taking the a' based on the current policy is the principle difference between Q-learning and SARSA.

SARSA versus Q-learning



Algorithm: SARSA

On each (s, a, r, s', a') :

$$\hat{Q}_{\pi}(s, a) \leftarrow (1 - \eta)\hat{Q}_{\pi}(s, a) + \eta(r + \gamma\hat{Q}_{\pi}(s', a'))$$



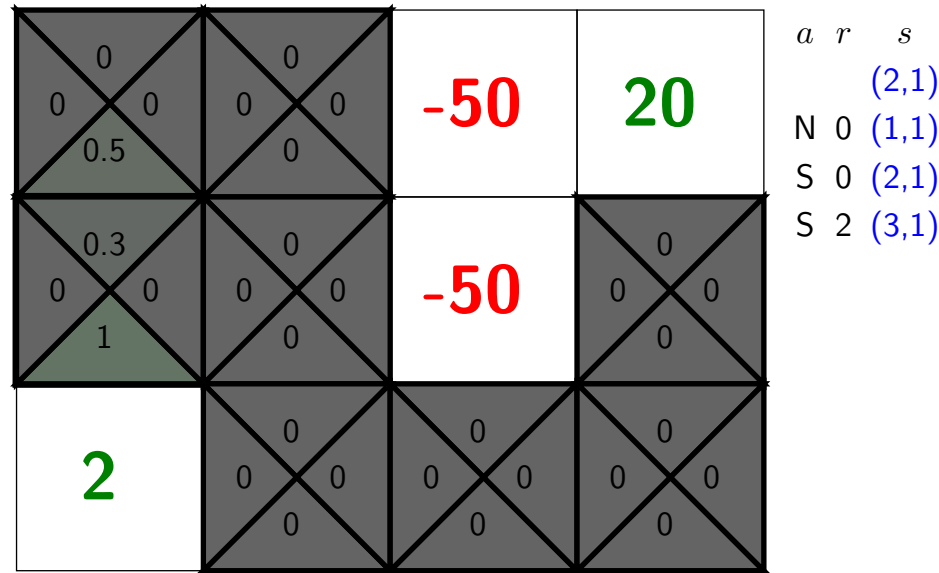
Algorithm: Q-learning [Watkins/Dayan, 1992]

On each (s, a, r, s') :

$$\hat{Q}_{\text{opt}}(s, a) \leftarrow (1 - \eta)\hat{Q}_{\text{opt}}(s, a) + \eta(r + \gamma \max_{a' \in \text{Actions}(s')} \hat{Q}_{\text{opt}}(s', a'))]$$

Volcanic SARSA and Q-learning

Run (or press ctrl-enter)



Utility: 2

- Let us try SARSA and Q-learning on the volcanic example.
- If you increase `numEpisodes` to 1000, SARSA will behave very much like model-free Monte Carlo, computing the value of the random policy.
- However, note that Q-learning is computing an estimate of $Q_{\text{opt}}(s, a)$, so the resulting Q-values will be very different. The average utility will not change since we are still following and being evaluated on the same random policy. This is an important point for **off-policy** methods: the online performance (average utility) is generally a lot worse and not representative of what the model has learned, which is captured in the estimated Q-values.

off policy vs on policy



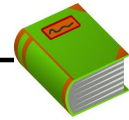
একটি policy follow করে data

collect, optimal policy (অন্য একটা)

estimate করার try করছি

Given policy এর different
property estimate

Off-Policy versus On-Policy



Definition: on-policy versus off-policy

On-policy: evaluate or improve the data-generating policy

Off-policy: evaluate or learn using data from another policy

policy evaluation

on-policy

off-policy

Monte Carlo
SARSA

policy optimization

Q-learning

- What do we mean by off-policy?
- Model-free Monte Carlo depends strongly on the policy π that is followed; after all it's computing Q_π . Because the value being computed is dependent on the policy used to generate the data, we call this an **on-policy** algorithm. In contrast, model-based value iteration is **off-policy**, because the model we estimated did not depend on the exact policy (as long as it was able to explore all (s, a) pairs).
- Further, model-free Q-learning is also **off-policy**, since it can learn the optimal policy using data from other policies.

Reinforcement Learning Algorithms

Algorithm	Estimating	Based on
Model-Based Monte Carlo	$\hat{T}, \hat{R}$	$s_0, a_1, r_1, s_1, \dots$
Model-Free Monte Carlo	$\hat{Q}_\pi$	u
SARSA	$\hat{Q}_\pi$	$r + \hat{Q}_\pi$
Q-Learning	$\hat{Q}_{\text{opt}}$	$r + \hat{Q}_{\text{opt}}$

out of syllabus

একটা policy follow করে data collect



একটা state এর next state, action same (overall iteration)

data not good enough

So ভালো optimal policy পাওয়ায়,



Soln! Exploration (Random action নিতে হয়)

↳ to better know the underlying environment

by formulation:

শুরু দিবে γ high

observation বাড়ান $\gamma \downarrow$

} γ challenge

শুরু দিবে observation ও বেশি weight দিচ্ছি। γ হলে শুরুর দিবে estimate খুব
poor. পরের দিবে better estimate. So মেমোরীকে বেশি weight দেয়া দরকার ছিল।

out of syllabus

η is definition $\rightarrow$ counter intuitive

$$(1-\eta) Q_{opt} \dots$$

$\uparrow$
model দিচ্ছে replace করে learn করার try করা হয়

